

CHAPTER V

MAIMONIDES AND THE SCIENTISTS

THE theologians are no danger because (as we have seen) they are wrong. The "scientists" (i.e., the Aristotelian philosophers) are a danger, and that because they are right, that is, right in what concerns their own province. Where the scientists go wrong is not in their chosen field of knowledge. In that they are sound, and on the whole reliable. Yet they often stray out of their own field and argue, or allow others to argue in their name, from their specialist knowledge of one part of the world to the nature of the world as a whole, or from the known nature of the world as it is now to the presumed nature of the world as it was supposedly once.

This procedure, says Maimonides, is illegitimate. It involves a fallacy (of the kind called many centuries later by Kant transcendental) because through it scientific thought leaves its proper sphere and intrudes into a sphere inappropriate to it. It can thus never lead to demonstration, only probability; its propositions are not "necessary" but only "admissible." Maimonides argues therefore, much as did Kant later, that such theories, being "admissible" only, not "necessary," offer no ground for rejecting faith.

As we have remarked before, science for the twelfth and thirteenth centuries meant Aristotle; and Maimonides' attitude to Aristotle is throughout of great interest. In general he may be said to accept his facts but to criticize his theories. He is occasionally critical of his facts as well; and he certainly understood that science, and in particular mathematical science, had not stood still with Aristotle. But in the broad he accepts the facts as described by the Aristotelian tradition; his problem is: what follows from them? How far is the scientific world-scheme reconcilable, or rather (and this is the peculiar Maimonidean approach) *not irreconcilable*, with the religious—Aristotle with the Bible? It must be emphasized again that Maimonides does not dream of suggesting the

rejection of science. On the contrary, it is on the facts assembled by science that he bases his own faith. But scientific facts are often used, even by the scientists themselves, in the interest of non-scientific conclusions, and Maimonides is on the watch for such illegitimate conclusions in so far as they conflict with religion. But even so he does not reject them; he only points out that they are inconclusive.

Maimonides' method *vis-à-vis* the scientists is therefore dialectical. His aim is to arrive at the conclusions which the theologians had failed to prove, on the basis established by science, and this object is achieved by insisting not that the former are demanded by the latter, but that they are not incompatible with it. If the facts of science (as opposed to the theories advanced by some scientists) allow the possibility of religion, and if religion has good claim to exist on other grounds, this claim may be upheld, and thus the way is made clear for the acceptance of a rational faith.

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The first step is to examine the attitude of science to the idea of God itself. Here Maimonides has an easy task since the Aristotelian tradition as systematized by the Arab philosophers accepted the existence, the incorporeality, and the unity, of God. God is the unitary source of all process and change, of all growth and decay, in the material universe; and being the source of change, is himself beyond change. But matter is the ground of change and division. God is therefore immaterial and one.

The essential part of the argumentation we have so briefly summarized is to be found in the text of Aristotle himself, and it is based upon a theory of the nature of motion which leads up to the proof of the existence of a "first mover" who is God. This God "moves" the universe ("as the beloved moves a lover") through the attraction exercised by his perfection. While the cause of all motion he is thus himself "unmoved"; and he is eternal, unchanging.

Maimonides, as I have said, has an easy task in taking over the arguments of the philosophers. But in a sense it is too

easy; the philosophers prove too much. If motion is eternal, then the universe which it fills is eternal too, and we are faced with the denial of creation. Maimonides is aware of this and says so at once. This great issue will be discussed very soon. But his object at the moment is to establish the point that the believer need not give up the idea of God from fear of the philosophers; on the contrary, it is the philosophers themselves who have provided proof of his existence. On the basis of established scientific principles, recapitulated by Maimonides in twenty-six concise propositions (II, Introd. and i-ii), the idea of God stands firm.

The same point emerges even more strikingly in the treatment of angels which follows immediately (iii-viii). The Bible mentions secondary spiritual beings who perform God's word and are his "messengers." Can we believe that such exist? Maimonides identifies them boldly with the "separate intelligences" which according to the doctrine of Aristotle accompany the "spheres" which form a kind of frame in which the stars are set and move, and which constitute the channel whereby motion is transferred through the various levels of the universe until it reaches our earth. The details of the mechanism are sufficiently obscure¹ and need not detain us here. The point to be seized is that Maimonides is seeking to justify religious belief by pointing out analogues in the accepted scientific account of the universe, and thus helping the believer who is perplexed at the strangeness of the Biblical narrative. The Bible is not strange, or at least not radically strange; it deals with the same objects and ideas as are found in the sciences, and comes to very similar conclusions, howbeit under different names.

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But the God of Aristotle, who is the principle of order in an already existing world, is not the God of the Bible who called all things into being from nothing, and there would

¹For Maimonides' own view see *Code I, Foundations*, ii-iii. The development of the theory (from Aristotle, *Metaphysics* Λ, viii-ix) can be followed in Duhem's great work referred to above, p. 56 n.

here seem to be a radical difference indeed. Eternity and free creation:—are these reconcilable by any means at all; and if they are not, which are we to sacrifice?

Maimonides is candid enough, in a sentence which was to bring down upon him all the wrath of Spinoza, to say that if the theory of "eternity" had really been demonstrated by the scientists, he would have found a way to get round the text of the Bible (II, xxv, 199). But the theory of eternity had not been demonstrated. The so-called demonstration rested on the "transcendental fallacy" (xv–xvii). We cannot argue from the world as it is now, to the world as it might once have been. Motion as it is now is eternal, but there was a time when there was no such thing; and if the scientist will counter with the argument that time is a form of motion, and that there cannot be a time before time, and that therefore time, and with it motion, is eternal, we can only reply that he is assuming what has to be proved: if time had a beginning, that beginning, although the beginning of time, was by the nature of the case not itself in time. We remain free therefore to discuss the alternative between eternity and creation on other grounds.

These grounds are both scientific and religious, and Maimonides sets the theories of mechanism and free creation one against the other. We are asked to consider the universe on the one side as a machine set up once and for all and now proceeding according to unvarying laws; on the other side, as a product of a free intelligence endowed with reason and will. And here Maimonides is helped by the very identification of angels with natural forces which would seem to have delivered him bound and captive to the mechanists. It is not so much that the angels are natural forces. Natural forces are angels; and, as Aristotle himself agreed, these angels or intelligences are living and self-conscious. The laws of nature are thus not iron necessity. They bear the attributes of self-conscious will. In fact nature and its laws *are* the divine will.

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Students of philosophy will remember very similar argumentation in Berkeley and many more modern thinkers as

well; and the debate still goes on. As for us, so for Maimonides the important point is to break down the hardness of the opposition between the two opposing theories. He will have no "fundamentalist" view, the view (II, xii, 179) that God literally commands and literally interferes with everything in the world, and is immediately active in all change. This view of the theologians, as we have seen, is wrong. God acts through secondary causes. And these secondary causes are regular: law is law, i.e., in theological phrase, God does not contradict himself. Indeed Maimonides goes further and insists that "will" and "wisdom" as applied to God are only "homonyms" and have a different meaning when they are applied to man. In searching for an expression to meet the view which seems to him the right one to convey the connection between God and the world, he finds it in the word "emanation" (II, xii), and speaks of continual and regular free creative activity passing through intermediaries till it irradiates all things and all men. He would seem to be aiming at a combination of the theories of immanence and transcendence, and to find it in the idea of design.

A special chapter (II, xix) takes up the argument with the scientists. After all they have never demonstrated the "why" of anything; they have only described the "what." They have never shown the reason for this, that or the other complication in the world, or indeed for there being any complication at all:

"What determined that the one small part should have ten stars, and the other portion should be without any star? And the whole body of the sphere being uniform throughout, why should a particular star occupy the one place and not another?" (188)

No doctrine of necessity can tell us that, only a belief in design. The very variety of the universe provides a reasonable argument for the existence of a free cause as its unitary ground.

Yet here again Maimonides is commendably cautious. The design of God is not the design of man, and the real meaning of the word in connection with God is a negative one; it

signifies the opposite of blind necessity. But blind necessity is not really required even by science. On the contrary, the notion of law as used in the sciences is commonly exaggerated and misunderstood. Facts (which must be accepted) are often confounded with theories (which need not). In astronomy, for example, the theories which are put forward as "laws" are not really laws at all. They are only descriptions of observed facts and their aim is to colligate as many facts as possible: they are hypotheses framed to "save the phenomena" (II, xi, 167; xxiv, 198). There is no reason why the theory of law and the theory of design should be held to conflict with one another. The will of God expresses its design in uniformities which it is the task of science to trace and register in what are called laws; but the laws are no more than the summary descriptions of the observed ways in which the will of God works in the world.

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Thus the theory of design is possible even from the point of view of science. Indeed it would seem to be necessary too, since without it we are not in a position to understand the observed variety of the universe. We may therefore turn with a clear conscience to another type of consideration—the religious one. The doctrine of design enables us to admit, what the doctrine of mechanism necessarily excludes, the idea of miracle. A free cause acting by design cannot be denied the theoretical possibility of interrupting the regular chain of causation it has itself initiated.

Miracles are thus not impossible; but having demonstrated this theoretically, Maimonides shows himself anxious to disembarass himself of them, as far as he can, in fact. As we have already seen, he suggests interpreting away the physical side of many Biblical miracles and relegating them to the world of visions. So far as they are held to have taken place as physical facts, he would seem willing to avail himself of a hint in Rabbinical literature to the effect that they were part of the pre-determined order of things; that, for example, God had arranged on the first day of creation that the Red Sea

should be divided before the Israelites and that the lions should not hurt Daniel (II, xxix, 210). The miracle forms as it were a part of the original "law"; the "law" as laid down contained provision for that particular exception.

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We need not delay long on this method of dealing with the problem of historic miracles which we find already in the *Commentary* (*Eight Chapters*, vii; *Aboth*, v). It would not seem to be more convincing than that of the Victorian geologist who thought that the world was created complete with aeons-old fossils. But there is another and more general point in Maimonides' discussion which is of especial interest and of permanent importance. While he is disposed to side with the religionist as against the scientist in asserting the *possibility* of miracle, he is quite as much disposed, with the scientist as against the religionist, to deny its exclusive *importance*. He says in effect that we do not believe in religion because we believe in miracles; we admit the possibility of miracle because we believe in religion. And this he lays down even in the *Code* (*Foundations*, viii, 1): "Moses our teacher was not believed in by the Israelites because of the miracles he wrought. One whose belief depends on miracles is of imperfect faith, since miracles can be wrought by magic and sleight-of-hand."

This doctrine is by no means new. It would seem to be implicit in the passage of Deuteronomy (xii, 2-4) from which it would appear that the successful working of miracles is not necessarily a sign of a *true* prophet, and the same line of thought is followed by one of Maimonides' great predecessors in the philosophy of religion, the Gaon Saadyah (*Emunoth* III, viii), who expresses astonishment at the idea that religion should be made to rest only on the evidence of miracles performed. And this brings us, as it brought Maimonides, to a consideration of one of the great miracles, the fact of prophecy, the existence of men who are men of God and bring the word of God to men.

Here, too, the special character of Maimonides' approach

must be stressed. He does not doubt the fact of revelation. The message brought by Moses is for him authentic; it is the veritable word of God. But the very notion of revelation is repugnant to naturalist thinkers, and the believer is therefore "perplexed." It is Maimonides' task therefore to show that when properly understood revelation is no unnatural phenomenon. Indeed, if the scientists were to understand the process involved in their own thinking, they would see that it itself is akin to prophecy.

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Among the various philosophical disciplines now recognized, along with the traditional ethics, logic and metaphysics, is that called epistemology or "theory of knowledge." As it exists at the present time it is largely a product of post-Kantian reflection on the Kantian question: "How can I know?" but in essence it is as old as systematic thought itself; indeed the best introduction to it has been declared to be the *Theaetetus* of Plato. What is to be said about Maimonides' account of prophecy (II, xxxii-xxxviii) may be perhaps most summarily expressed by saying that it is his theory of knowledge.

Historically speaking, the account goes back in essence to Plato. It makes much use of Aristotle, particularly of one obscure passage in his book on psychology, the *De Anima*; and in relation to the religious phenomenon of prophecy it was worked out by certain Arab philosophers—more especially by Avicenna. Without entering into technicalities or endeavouring to trace the historical stages of its development, I shall try and set down its main points.

There are three principal stages in the development of human understanding, and corresponding with them there are three different types or ranges of objects distinguished, and three different levels of mental powers involved. After the first shadowy gropings of an infant there begin to emerge in his consciousness definite things. These are apprehended as individual images, and as such they constitute the world of the ordinary man. As thinking proceeds, the things cease to be seen as individual images. They begin to be grasped, in their

similarity and dissimilarity with others, as representatives of classes or types, not as particular things doing this or that particular action or exhibiting this or that particular feature. Thus there emerge general relations or laws. This is the stage of science and the scientific intellect; and it is best represented by the mathematician as he perceives the widest possible relations in number and space, and expresses them in formulæ which are true universally and of necessity. Yet, as Aristotle insisted, although science is of the universal, all substance is individual: general formulæ will not enable us to grasp concrete things. In order to grasp concrete things (not as mere perceived images which may by chance have passed before the retina of our eyes—this is mere accident and not knowledge—but as rooted in the structure of the universe), we need a fusion at a higher degree of the first and second stages so that the particularity of the one can appear together with the necessity of the other.

This fusion Maimonides found in the phenomenon of prophecy. The first stage is that of the "imagination" (i.e. the grasping as mere image); the second is that of the "intellect" (called "discursive" because it goes from point to point and is only enabled to generalize by ignoring individual characteristics). In order to penetrate to the essence of things, or rather, in order to grasp things in their essence, we thus need "imaginative intellect" or "intellectual imagination," the power which is often referred to in modern times as intuition. This intuition is for Maimonides what constitutes prophecy. True, it may be withheld by the will of God; but when it appears this is its nature.

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An important point must be added at once, and here the obscure phrase of Aristotle became central. All knowledge is from above, and prophecy, far from being an exception, is in a sense the rule. For we must ask, on the ordinary ground of the theory of knowledge, how the mind begins to work at all. Following Aristotle, we may take a plain example from sense-perception. We have eyes, but we do not actually see until an external object passes before them; till then we have only the

capacity of sight. The same holds at every stage of our knowing. Until something comes to stimulate it to activity, mind is only potential. Potentially it is all that it can ever be. Actually it is nothing. It is an unlit match, an unwound spring; and if it were not for an activating force, it would remain passive and we would never think at all.

This is the specific ground given by Maimonides (II, iv, 158) as sufficient to prove the existence of the "separable" intellect (which is "activity" and "makes all things") mentioned by Aristotle (*De Anima* III, v), and such indeed is Aristotle's own ground for the assumption.¹ It is this "active intellect" which sets our minds moving, and causes them to change their mere capacity to think into actual thinking. The Active Intellect is all knowledge in an actual state. Thus the more knowledge we acquire, i.e. the more we actualize our minds, the more we approximate to perfect reason.

Clearly the Active Intellect is one since knowledge is one; it is also non-human and non-corporeal since it is common to all. But whether it be God, as some commentators held, or a higher part of our own minds (yet common to all minds), as another school of commentators supposed, is a question difficult to answer. Maimonides takes a middle way² and asserts it to be the lowest of the order of the Intelligences (or angels) mediating between the "unmoved mover" and the world of man. It is pure intellect calling out ours.

Thus the knowledge of the prophet is only the highest type of knowledge. All knowledge is ultimately revelation. Without an awakening from above there would be not only no prophecy; there would be no knowledge at all. We would be passive receivers of images with no more knowledge than a mirror or the waters of a brook; the shadows would flit over our consciousness (if we could be said to have consciousness) and never stay. Knowledge consists in the grasping of the universal in the particular, and this is an active process; but an active process needs activation, and all activation goes back

¹ See text, and cf. R. D. Hicks' Introduction, p. lxiii. On the history of the theory in the middle ages, see G. Théry, *Alexandre d' Aphrodise*, Kain, 1926. Its affinities with the all-important Augustinian "illuminism" of the sixteenth and seventeenth centuries should be noted.

² Cf. G. R. G. Mure, *Aristotle* (Oxford, 1932), p. 175.

ultimately to one source. Thus the influence which streams from God does not move only the physical machinery of the universe. It is the source too of our higher mental life. The divine mind is the active, energizing pattern of all our thinking which is thus seen to be almost literally inspired.

Whatever one may think of this theory (and it exercised profound influence on all types of speculation both in the Middle Ages and long after), Maimonides' use of it is clear. It enables him to set the phenomenon of prophecy within an intelligible world scheme. The perplexed need not be disturbed by the disdain of the scientist. The scientist himself needs some such theory to explain his own science. The point is much the same as often found in our own contemporary literature, only our antithesis to science is not theology but art. So we tell our poets they need not fear to be laughed at by scientists since science too needs the creative vision; indeed without the constructive imagination which is the poetic gift, there would be no science at all.

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What seems to us to be right and most fruitful in Maimonides' theory of prophecy is his stress on the element of imagination, the strong pictorial power which enabled prophets to grasp and present moral ideas, as they work in the fates of individual man or whole empires, in an unforgettable parable or symbol: "the little ewe lamb"; "all flesh is grass"; "Assyria the rod of my anger." But what struck his contemporaries most was Maimonides' calm assumption that the prophets were scientists; and necessarily so, since prophecy represents that highest stage of thinking in which science and imagination are fused. And we too may be puzzled. Yet gratitude is due to Maimonides at least in so far as he freed the prophets from the swaddling-clothes of magic and soothsaying, and placed them within the orbit of reason. We may not like to think of Isaiah and Jeremiah as students of the natural sciences; but they should be conceded to have been at least thoughtful men, not mere showy necromancers and diviners of what might happen

to-morrow, petty gipsies in the world's fair. Maimonides makes it clear he does not look on them as scientists only; indeed the mere scientist is no more a scientist for him than for Pascal. And if the scientist is indeed touched with inspiration from on high, he would seem to be no unworthy yoke-fellow to the prophet; while it was a prophet who said, referring to the witness in the skies: "Lift up your eyes on high and see who hath created these."